

A geometric mean beats the logarithm

$$\int_k^{k+1} \frac{dx}{x} < \frac{1}{\sqrt{k(k+1)}}, \text{ term by term}$$

$$\boxed{\frac{1}{\sqrt{1^2+1}} + \frac{1}{\sqrt{2^2+2}} + \cdots + \frac{1}{\sqrt{n^2+n}} > \log(n+1), \quad n \geq 1.}$$

Overview. Both sides telescope into n matching pieces. On the right,

$$\log(n+1) = \sum_{k=1}^n \log \frac{k+1}{k} = \sum_{k=1}^n \int_k^{k+1} \frac{dx}{x},$$

and on the left the k -th term is the integral of the constant $\frac{1}{m}$ over the same interval, where

$$m = \sqrt{k(k+1)} = \sqrt{k^2+k}$$

is the geometric mean of the endpoints. So the whole inequality follows from the single statement

$$\int_k^{k+1} \frac{dx}{x} < \frac{1}{m}, \quad m = \sqrt{k(k+1)}. \quad (*)$$

This is a strict comparison between the average of $\frac{1}{x}$ on $[k, k+1]$ and its value at the geometric mean of the endpoints. Convexity is not enough: it compares with the *arithmetic* mean and yields the inequality in the useless direction. It follows instead from the reflection symmetry $x \mapsto m^2/x$, which is exactly the symmetry fixing the geometric mean.

Two proofs of $(*)$ are given: one by the reflection, and one by a change of variable that reduces it to the classical $2 \log s < s - \frac{1}{s}$.

1. Reduction to a single interval

Telescoping the logarithm,

$$\log(n+1) = \log \frac{2}{1} + \log \frac{3}{2} + \cdots + \log \frac{n+1}{n} = \sum_{k=1}^n \int_k^{k+1} \frac{dx}{x}. \quad (1)$$

Since $m = \sqrt{k(k+1)}$ is a constant on $[k, k+1]$,

$$\frac{1}{\sqrt{k^2+k}} = \int_k^{k+1} \frac{dx}{m}. \quad (2)$$

So the difference of the two sides of the problem is

$$\sum_{k=1}^n \int_k^{k+1} \left(\frac{1}{m} - \frac{1}{x} \right) dx,$$

and it suffices to prove that each summand is positive, i.e. $(*)$.

2. First proof: reflect about the geometric mean

Fix $k \geq 1$ and $m = \sqrt{k(k+1)}$. By AM–GM, $k < m < k+1$, so the integrand $\frac{1}{m} - \frac{1}{x}$ changes sign exactly once, at $x = m$: it is negative on $[k, m)$ and positive on $(m, k+1]$. Write

$$\int_k^{k+1} \left(\frac{1}{m} - \frac{1}{x} \right) dx = - \underbrace{\int_k^m \left(\frac{1}{x} - \frac{1}{m} \right) dx}_{S_-} + \underbrace{\int_m^{k+1} \left(\frac{1}{m} - \frac{1}{x} \right) dx}_{S_+}, \quad (3)$$

with $S_{\pm} > 0$. The claim is $S_+ > S_-$.

The decisive observation is that the involution

$$x \mapsto \frac{m^2}{x}$$

maps $[k, m]$ onto $[m, k+1]$, because $\frac{m^2}{k} = \frac{k(k+1)}{k} = k+1$. This is the only place where the choice $m = \sqrt{k(k+1)}$ is used, and it is the reason that choice is forced.

Substituting $x = m^2/t$, $dx = -\frac{m^2}{t^2} dt$, in S_- :

$$\left(\frac{1}{x} - \frac{1}{m} \right) dx = \left(\frac{t}{m^2} - \frac{1}{m} \right) \left(-\frac{m^2}{t^2} \right) dt \quad \longrightarrow \quad S_- = \int_m^{k+1} \frac{t-m}{t^2} dt. \quad (4)$$

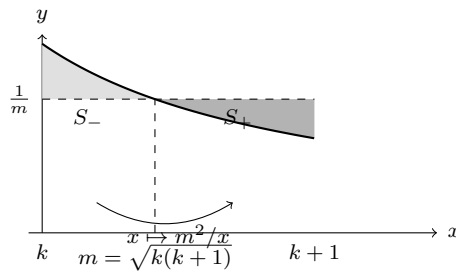
Meanwhile

$$S_+ = \int_m^{k+1} \frac{x-m}{mx} dx. \quad (5)$$

Both integrals are now over the *same* interval, so they can be subtracted pointwise:

$$S_+ - S_- = \int_m^{k+1} (x-m) \left(\frac{1}{mx} - \frac{1}{x^2} \right) dx = \int_m^{k+1} \frac{(x-m)^2}{m x^2} dx > 0. \quad (6)$$

The integrand is a perfect square divided by positive quantities, and it is not identically zero. Hence (*) holds strictly.



3. Second proof: one substitution

The inequality (*) can also be normalised away entirely. Put

$$s = \sqrt{\frac{k+1}{k}} > 1.$$

Then

$$\int_k^{k+1} \frac{dx}{x} = \log \frac{k+1}{k} = 2 \log s,$$

and, since $\frac{1}{k} = s^2 - 1$ and $\sqrt{k(k+1)} = ks$,

$$\frac{1}{\sqrt{k(k+1)}} = \frac{1}{ks} = \frac{s^2 - 1}{s} = s - \frac{1}{s}.$$

So $(*)$ is precisely

$$\boxed{2 \log s < s - \frac{1}{s} \quad (s > 1).} \quad (7)$$

This is immediate: let $h(s) = s - \frac{1}{s} - 2 \log s$. Then $h(1) = 0$ and

$$h'(s) = 1 + \frac{1}{s^2} - \frac{2}{s} = \left(1 - \frac{1}{s}\right)^2 \geq 0,$$

with strict positivity for $s > 1$. Hence $h(s) > 0$ for $s > 1$. $\square$

Inequality (7) has independent interest. It states that $\log$ grows more slowly than the odd hyperbolic-type function $s \mapsto \frac{1}{2}(s - s^{-1})$, with contact of order three at $s = 1$. Written with $s = e^{u/2}$ it reads $u < 2 \sinh \frac{u}{2}$, which is the familiar $\sinh t > t$.

4. Conclusion and sharpness

Summing $(*)$ over $k = 1, \dots, n$ and using (1) and (2),

$$\log(n+1) = \sum_{k=1}^n \int_k^{k+1} \frac{dx}{x} < \sum_{k=1}^n \frac{1}{\sqrt{k^2 + k}},$$

which is the assertion.

The inequality is not far from an equality in the following sense. From (6) the k -th gap is

$$\frac{1}{\sqrt{k^2 + k}} - \log \frac{k+1}{k} = \int_m^{k+1} \frac{(x-m)^2}{mx^2} dx = \frac{1}{24k^3} + O\left(\frac{1}{k^4}\right),$$

as one checks by expanding both terms in $\varepsilon = \frac{1}{k}$: $\frac{1}{\sqrt{k^2 + k}} = \varepsilon - \frac{\varepsilon^2}{2} + \frac{3\varepsilon^3}{8} - \dots$ and $\log(1+\varepsilon) = \varepsilon - \frac{\varepsilon^2}{2} + \frac{\varepsilon^3}{3} - \dots$, whose difference starts at $(\frac{3}{8} - \frac{1}{3})\varepsilon^3$. So the two sides differ by a convergent series and

$$\sum_{k=1}^n \frac{1}{\sqrt{k^2 + k}} - \log(n+1) \longrightarrow c = 0.0190285404\dots$$

The excess is a genuine constant, not a growing quantity: the inequality is asymptotically tight at the level of the leading term $\log n$.

n	$\sum_{k \leq n} (k^2 + k)^{-1/2}$	$\log(n+1)$	difference
1	0.7071068	0.6931472	0.0139596
5	1.8102110	1.7917595	0.0184517
50	3.9508465	3.9318256	0.0190205
1000	6.9277834	6.9087548	0.0190285

5. Why the geometric mean, and what fails otherwise

The natural first attempt is to bound $\int_k^{k+1} \frac{dx}{x}$ using convexity of $\frac{1}{x}$. The midpoint rule for a convex function gives

$$\int_k^{k+1} \frac{dx}{x} \geq \frac{1}{k + \frac{1}{2}},$$

the wrong direction, and indeed $\frac{1}{k+1/2} < \frac{1}{\sqrt{k^2+k}}$ since $k + \frac{1}{2} > \sqrt{k(k+1)}$ by AM–GM. Convexity alone is therefore not enough; one needs the specific point at which the reflection symmetry of $\frac{1}{x}$ is centred.

Comparison point $c \in [k, k+1]$	$\int_k^{k+1} \frac{dx}{x}$ vs $\frac{1}{c}$	Reason
$c = k + \frac{1}{2}$ (arithmetic mean)	$\int > \frac{1}{c}$	Midpoint rule for a convex function.
$c = \sqrt{k(k+1)}$ (geometric mean)	$\int < \frac{1}{c}$	Sections 2–3; the map $x \mapsto c^2/x$ exchanges the two halves.
$c = \frac{1}{\log \frac{k+1}{k}}$ (logarithmic mean)	equality	By definition of the logarithmic mean.

The three rows are the classical chain

$$\text{harmonic} \leq \text{geometric} \leq \text{logarithmic} \leq \text{arithmetic},$$

and the problem is exactly the second inequality of that chain applied to the pair $(k, k+1)$. Seen this way, $(*)$ is a standard inequality with a name, and the reflection proof of Section 2 is the usual proof of $G \leq L$.